

Review Session 1

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Theorem 1 (Multivariate Wold Decomposition). *Let $\{X_t\}_{t \in \mathbb{Z}}$ be a mean-zero, covariance-stationary, $\mathbb{R}^n$ -valued stochastic process with finite second moments. Then there exists a vector-valued white-noise process $\{\varepsilon_t\}_{t \in \mathbb{Z}}$ with $\mathbb{E}(\varepsilon_t \varepsilon_t') = \Sigma_\varepsilon$ positive definite and a sequence of matrices $\{\Theta_k\}_{k \geq 0}$ satisfying*

$$\sum_{k=0}^{\infty} \text{tr}(\Theta_k \Sigma_\varepsilon \Theta_k') < \infty$$

such that

$$X_t = \sum_{k=0}^{\infty} \Theta_k \varepsilon_{t-k} + D_t,$$

where D_t is orthogonal to the closed linear span of $\{\varepsilon_s : s \leq t\}$. If $\{X_t\}$ is purely non-deterministic, then $D_t = 0$ and the representation reduces to a (one-sided) moving-average representation.

Proof. Let $\mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$ be a space of square integrable random vectors. In other words, we have

$$\mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P}) = \left\{ X : \Omega \rightarrow \mathbb{R}^n \left| \int_{\Omega} X(\omega)' X(\omega) \mathbb{P}(d\omega) < \infty \right. \right\}$$

where the set Ω is a sample space, the collection of sets $\mathcal{F}$ is a sigma field, and the set function $\mathbb{P} : \mathcal{F} \rightarrow \mathbb{R}$ is a probability measure. The space $\mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$ is Hilbert space if we introduce an inner product

$$\langle X, Y \rangle = \int_{\Omega} X(w)' Y(w) \mathbb{P}(dw) = \mathbb{E}[X'Y]$$

for all $X, Y \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$. The random vector $X \in \mathbb{R}^n$ defines a mapping from the measure space $(\Omega, \mathcal{F}, \mathbb{P})$ to $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n), P_X)$ where $P_X : \mathcal{B}(\mathbb{R}^n) \rightarrow \mathbb{R}$ is the probability measure induced by X defined as

$$P_X(A) = \mathbb{P}\{X^{-1}(A)\} = \mathbb{P}\{w | X(w) \in A\}$$

for all $A \in \mathcal{B}(\mathbb{R}^n)$. Let $\{X_s\}_{s=-\infty}^t \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$. For each t , define the closed linear subspace generated by the past as $\mathcal{M}_t = \text{span}(\{X_s\}_{s=-\infty}^t)$, i.e.,

$$\mathcal{M}_t = \left\{ \sum_{s=-\infty}^t A_s X_s : A_s \in \mathbb{R}^{n \times n} \right\}$$

This means that an element $Y \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$ is in $\mathcal{M}_t$ if

$$Y = \sum_{s=-\infty}^t A_s X_s$$

in norm, i.e., $\mathbb{E} \left(Y - \sum_{s=-\infty}^t A_s X_s \right)' \left(Y - \sum_{s=-\infty}^t A_s X_s \right) = 0$, for some $(A_s)_{s=-\infty}^t$.

Let P_{t-1} denote the orthogonal projection onto $\mathcal{M}_{t-1}$ and define the innovation

$$\varepsilon_t = X_t - P_{t-1} X_t.$$

By construction,

$$\varepsilon_t \in \mathcal{M}_t, \quad \varepsilon_t \perp \mathcal{M}_{t-1}.$$

Since $\mathcal{M}_s \subset \mathcal{M}_{t-1}$ for all $s < t$, it follows that

$$\mathbb{E}(\varepsilon_t \varepsilon_s') = 0, \quad s < t,$$

so $\{\varepsilon_t\}$ is a vector-valued white-noise sequence in L^2 .

Define the closed linear span of present and past innovations by

$$\mathcal{E}_t = \text{span}\{\varepsilon_t, \varepsilon_{t-1}, \varepsilon_{t-2}, \dots\} \subset \mathcal{M}_t.$$

Project X_t onto $\mathcal{E}_t$ and write

$$X_t = \sum_{k=0}^{\infty} \Theta_k \varepsilon_{t-k} + D_t,$$

where $D_t \perp \mathcal{E}_t$ and the series converges in L^2 .

Because the innovations are mutually orthogonal, each coefficient Θ_k is determined uniquely from the minimization problem

$$\min_{\Theta_k} \|X_t - \Theta_k \varepsilon_{t-k}\|_{L^2}^2 = \min_{\Theta_k} \mathbb{E} [(X_t - \Theta_k \varepsilon_{t-k})' (X_t - \Theta_k \varepsilon_{t-k})]$$

with the orthogonality condition

$$\mathbb{E}[(X_t - \Theta_k \varepsilon_{t-k}) \varepsilon_{t-k}'] = 0,$$

which yields

$$\Theta_k = \mathbb{E}(X_t \varepsilon_{t-k}') [\mathbb{E}(\varepsilon_{t-k} \varepsilon_{t-k}')]^{-1}.$$

Since the partial sum process $\sum_{k=0}^K \Theta_k \varepsilon_{t-k}$ must converge in L^2 as $K \rightarrow \infty$, we have

$$\sum_{k=0}^{\infty} \text{tr}(\Theta_k \Sigma_{\varepsilon} \Theta_k') < \infty.$$

If $\mathcal{M}_t = \mathcal{E}_t$ (pure nondeterminism), then $D_t = 0$, completing the proof. □

1 Invertibility of MA, AR process

In a very specific context of time series modeling of AR and MA models, the terminology 'causality' and 'invertibility' is defined as below. Suppose we have a vector moving average (VMA) model given by

$$X_t = \sum_{l=-\infty}^{\infty} \Theta_l \varepsilon_{t-l} \quad (1)$$

where we assume $\{\Theta_l\}_l$ is square summable and ε_t is orthogonal white noise. Assuming that X_t has been centered, the variance of X_t is given by

$$\begin{aligned} \mathbb{E}X_t X_t' &= \mathbb{E} \left(\sum_{l=-\infty}^{\infty} \Theta_l \varepsilon_{t-l} \right) \left(\sum_{m=-\infty}^{\infty} \Theta_m \varepsilon_{t-m} \right)' \\ &= \sum_{l=-\infty}^{\infty} \Theta_l \mathbb{E}[\varepsilon_{t-l} \varepsilon_{t-l}'] \Theta_l' \\ &= \sum_{l=-\infty}^{\infty} \Theta_l \Sigma \Theta_l' \end{aligned}$$

where we've used orthogonality of ε_t in the second equality. We assume that $\{\Theta_l\}_l$ is square summable, i.e.,

$$\sum_{l=-\infty}^{\infty} \|\Theta_l\|_F^2 = \sum_{l=-\infty}^{\infty} \text{trace}(\Theta_l' \Theta_l) < \infty$$

which ensures that X_t as defined by (1) is well-defined and is an element in $\mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$. We define the terminologies

1. Causality:

$$\mathbb{E}[\varepsilon_{t+k} X_t'] = 0$$

for all $k > 0$. This implies that

$$\Theta_k = 0, \forall k < 0$$

i.e.,

$$X_t = \sum_{l=0}^{\infty} \Theta_l \varepsilon_{t-l}$$

2. Invertibility (of ε_t with respect to X_T):

$$\varepsilon_t \in \text{span}(\{X_s\}_{s=-\infty}^t)$$

Suppose we have a general VAR model

$$X_t = \sum_{l=0}^{\infty} A_l X_{t-l} + \varepsilon_t$$

where ε_t is an orthogonal white noise. Then, we use the terminology

3. Invertibility (of X_t with respect to ε_t):

$$X_t \in \text{span}(\{\varepsilon_s\}_{s=-\infty}^t)$$

This is the causality of the VMA representation of the VAR model.

1.1 MA model

Let MA(q) model be given as

$$\begin{aligned} X_t &= \sum_{l=0}^q \Theta_l \varepsilon_{t-l} \\ &= \Theta_0 \varepsilon_t + \Theta_1 \varepsilon_{t-1} + \dots + \Theta_q \varepsilon_{t-q} \end{aligned} \tag{2}$$

It is not difficult to check that MA models are always (covariance) stationary. Note that timing indeterminacy exists in MA models, i.e., the model (2) has exactly the same second moments as

$$\begin{aligned} X_t &= \sum_{l=0}^q \Theta_l \varepsilon_{t-k-l} \\ &= \Theta_0 \varepsilon_{t-k} + \Theta_1 \varepsilon_{t-k-1} + \dots + \Theta_q \varepsilon_{t-k-q} \end{aligned}$$

for any choice of integer-valued k . This means that the data cannot tell us which MA model is better. To fix this indeterminacy we usually just assume $k = 0$. This implies that

1. MA model is causal
2. Today's shock ε_t starts to affect the observations today X_t .

This may not be desirable if we want to identify news shock.

1.2 AR model

A general VAR(p) model is defined as

$$X_t = \sum_{l=1}^p A_l X_{t-l} + \varepsilon_t$$

For the investigation of stationarity and invertibility of AR models, let us first consider a simple AR(1).

$$X_t = aX_{t-1} + \varepsilon_t$$

1. Suppose $|a| < 1$: By repeated substitution, we obtain

$$\begin{aligned}
X_t &= aX_{t-1} + \varepsilon_t \\
&= \varepsilon_t + a(\varepsilon_{t-1} + aX_{t-2}) \\
&\vdots \\
&= \sum_{l=0}^{T-1} a^l \varepsilon_{t-l} + a^T X_{t-T} \\
&\vdots \\
&= \sum_{l=0}^{\infty} a^l \varepsilon_{t-l}
\end{aligned}$$

whenever we have ' ∞ ' in our equation, we have to ask 'In what sense does the equality hold?'. The equality is used to denote that

$$\sum_{l=0}^{T-1} a^l \varepsilon_{t-l} + a^T X_{t-T} \rightarrow \sum_{l=0}^{\infty} a^l \varepsilon_{t-l}$$

in L^2 as $T \rightarrow \infty$. To check this, notice that

$$\begin{aligned}
\left\| \sum_{l=0}^{T-1} a^l \varepsilon_{t-l} + a^T X_{t-T} - \sum_{l=0}^{\infty} a^l \varepsilon_{t-l} \right\|_{L^2} &= \left\| a^T X_{t-T} - \sum_{l=T}^{\infty} a^l \varepsilon_{t-l} \right\|_{L^2} \\
&\leq a^T \|X_{t-T}\|_{L^2} + \left\| \sum_{l=T}^{\infty} a^l \varepsilon_{t-l} \right\|_{L^2} \\
&\rightarrow 0
\end{aligned}$$

as $T \rightarrow \infty$ since $X_t \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$ for all t and $\sum_{l=T}^{\infty} a^l \varepsilon_{t-l} \rightarrow 0$ as $T \rightarrow \infty$.

2. Suppose $|a| > 1$: Repeated substitution as in the case above does not give us a solution in $\mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$. However, we may write

$$\frac{1}{a}X_t = X_{t-1} + \frac{1}{a}\varepsilon_t$$

so that

$$\begin{aligned}
X_{t-1} &= \frac{1}{a}X_t - \frac{1}{a}\varepsilon_t \\
&= \frac{1}{a} \left(\frac{1}{a}X_{t+1} - \frac{1}{a}\varepsilon_{t+1} \right) - \frac{1}{a}\varepsilon_t \\
&= -\frac{1}{a}\varepsilon_t - \frac{1}{a^2}\varepsilon_{t+1} + \frac{1}{a^2} \left(\frac{1}{a}X_{t+1} - \frac{1}{a}\varepsilon_{t+1} \right) \\
&\vdots \\
&= -\sum_{l=0}^{\infty} \frac{1}{a^{l+1}} \varepsilon_{t+l}
\end{aligned}$$

(Please check that the last equation is well-defined). Thus, we have

$$X_t = - \sum_{l=1}^{\infty} \frac{1}{a^l} \varepsilon_{t+l}$$

This means that we can 'invert' X_t even if $|a| > 1$. However, the resulting solution is not a causal MA model and we call such an AR process not invertible.

3. $|a| = 1$: No stationary solution.

2 How to check invertibility for general AR or MA processes

We begin with the following two facts

1. (Taylor Expansion) Let f be a function $f : \mathbb{C} \rightarrow \mathbb{C}$. Suppose that f is analytic on an open set $D = \{z : |z - z_0| < \delta\}$ for some z_0 . Then we have

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n$$

for all $z \in D$.

2. (Laurent Expansion) If $f(z)$ is analytic in an annulus containing $C = \{z : r < |z - z_0| < R\}$ for some z_0 , we have

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n$$

for all $z \in C$ and for some $\{a_n\}_{n=-\infty}^{\infty}$. Furthermore, a Laurent expansion of a function is unique whenever it exists.

Suppose f is analytic on an open set containing unit disk $D = \{z : |z| \leq 1\}$. Then, using Taylor expansion at $z_0 = 0$, we have

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} z^n$$

for all $z \in D$. If we have singularities in the interior of the unit disk, we can use Laurent expansion. Suppose f is analytic in an open set containing unit circle $C = \{z : |z| = 1\}$. Then, using Laurent expansion at $z_0 = 0$, we have

$$f(z) = \sum_{n=-\infty}^{\infty} a_n z^n$$

for some (a_n) and for all $z \in C$. Since Laurent expansion is unique, Taylor expansion is Laurent expansion whenever it exists. Examples:

1. Suppose we have $f(z) = \frac{1}{1-z}$. Then, $f(z)$ is analytic for all $|z| < 1$. Thus, the Taylor series exists and is given by

$$f(z) = 1 + z + z^2 + \dots$$

2. Suppose we have $f(z) = \frac{1}{z(1-z)}$. Then, $f(z)$ is analytic for all $|z| < 1, z \neq 0$. Thus, the Laurent series exists and is given by

$$f(z) = \frac{1}{z} + 1 + z + z^2 + \dots$$

Suppose $f(z)$ is analytic in an open set containing the unit circle. Then, $f(z)$ admits the Laurent series given in the form

$$f(z) = \sum_{n=-\infty}^{\infty} a_n z^n \quad (3)$$

which is valid for all z in an open set containing the unit circle. We can further show that the Laurent coefficients are absolutely summable, i.e., $\sum_{n=-\infty}^{\infty} |a_n| < \infty$. To show absolute summability, note that we may take $\epsilon > 0$ small so that (3) is satisfied for all $1 - \epsilon < |z| < 1 + \epsilon$. This implies that

$$f(1 + \epsilon/2) = \sum_{n=-\infty}^{\infty} a_n (1 + \epsilon/2)^n < \infty$$

which gives us $a_n (1 + \epsilon/2)^n \rightarrow 0$ as $n \rightarrow \infty$. Thus, there exists $K > 0$ such that

$$|a_n| < K(1 + \epsilon/2)^{-n}, \quad \forall n = 0, 1, 2, \dots$$

Summing over $n \geq 0$, we have

$$\sum_{n=0}^{\infty} |a_n| < \infty$$

Similarly, we have

$$f(1 - \epsilon/2) = \sum_{n=-\infty}^{\infty} a_n (1 - \epsilon/2)^n < \infty$$

which implies $a_{-n} (1 - \epsilon/2)^{-n} \rightarrow 0$ as $n \rightarrow \infty$ and thus there exists $K > 0$ such that

$$|a_{-n}| < K(1 - \epsilon/2)^n, \quad \forall n = 1, 2, \dots$$

Summing over $n \geq 1$, we have

$$\sum_{n=-\infty}^{-1} |a_n| < \infty$$

Together, we obtain $\sum_{n=-\infty}^{\infty} |a_n| < \infty$.

2.1 AR(p)

Consider an AP(p) model given by

$$X_t = \sum_{l=1}^p \alpha_l X_{t-l} + \varepsilon_t$$

Let $\mathcal{M} = \text{span}(\{X_t\}_{t=-\infty}^{\infty})$. Define an operator $L : \mathcal{M} \rightarrow \mathcal{M}$ such that

$$LX_t = X_{t-1}$$

Then, we may write

$$\left(I - \sum_{l=1}^p \alpha_l L^l \right) X_t = \alpha(L) X_t = \varepsilon_t$$

where $\alpha(L) = I - \sum_{l=1}^p \alpha_l L^l$ is a linear filter and I is an identity operator. $\alpha(L)$ is also an operator from $\mathcal{M}$ to $\mathcal{M}$. We want to know whether the inverse operator $\alpha(L)^{-1}$ exists. If $\alpha(L)^{-1}$ exists, we can operate it on both sides of

$$\alpha(L) X_t = \varepsilon_t$$

so that we have

$$X_t = \alpha(L)^{-1} \varepsilon_t$$

Consider a function $\alpha : \mathbb{C} \rightarrow \mathbb{C}$ given by

$$\alpha(z) = 1 - \sum_{l=1}^p \alpha_l z^l$$

Note that even if we use the same letter α to denote two functions $\alpha(L)$ and $\alpha(z)$, these two functions are different functions with different domains and codomains. $\alpha(z)^{-1}$ is well defined as

$$\alpha(z)^{-1} = \frac{1}{1 - \sum_{l=1}^p \alpha_l z^l}$$

whenever the denominator does not equal zero. Suppose the denominator $\alpha(z) = 1 - \sum_{l=1}^p \alpha_l z^l$ does not have roots in an open set containing unit circle $C = \{z : |z| = 1\}$. Then, $\alpha(z)^{-1}$ is analytic in the open set, and we have

$$\alpha(z)^{-1} = \sum_{n=-\infty}^{\infty} a_n z^n$$

for some (a_n) . Note that we have

$$\alpha(z)^{-1} \alpha(z) = \left(\sum_{n=-\infty}^{\infty} a_n z^n \right) \left(1 - \sum_{l=1}^p \alpha_l z^l \right) = 1 \quad (4)$$

for all z in an open set containing a unit circle. If we write

$$\left(\sum_{n=-\infty}^{\infty} a_n z^n \right) \left(1 - \sum_{l=1}^p \alpha_l z^l \right) = \sum_{n=-\infty}^{\infty} c_n z^n$$

we immediately see that all the coefficients c_n on the right hand side with $n \neq 0$ is zero since (4) holds for a continuum of z . This implies that if $\alpha(L)^{-1}$ does exist, it should be given in the form

$$\alpha(L)^{-1} = \sum_{n=-\infty}^{\infty} a_n L^n$$

with (a_n) given by the Laurent expansion of $\alpha(z)^{-1}$. We finally need to check that $\alpha(L)^{-1}$ does exist, i.e., it is an operator from $\mathcal{M}$ to $\mathcal{M}$. It suffices to show that the filtered random variable Y_t given by

$$Y_t = \sum_{n=-\infty}^{\infty} a_n X_{t-n}$$

is square integrable. Note that we have

$$\begin{aligned} \mathbb{E} Y_t \bar{Y}_t &= \mathbb{E} \left(\sum_{n=-\infty}^{\infty} a_n X_{t-n} \right) \left(\sum_{m=-\infty}^{\infty} \bar{a}_m \bar{X}_{t-m} \right) \\ &= \sum_{n,m=-\infty}^{\infty} a_n \bar{a}_m \mathbb{E} X_{t-n} \bar{X}_{t-m} \\ &= (\max_{n,m} \mathbb{E} X_{t-n} \bar{X}_{t-m}) \sum_{n,m=-\infty}^{\infty} a_n \bar{a}_m \\ &= (\max_{n,m} \mathbb{E} X_{t-n} \bar{X}_{t-m}) \left(\sum_{n=-\infty}^{\infty} a_n \right) \left(\sum_{m=-\infty}^{\infty} \bar{a}_m \right) \\ &< \infty \end{aligned}$$

because $\mathbb{E}|X_t|^2 < \infty$ and $\sum_{n=-\infty}^{\infty} |a_n| < \infty$. Thus, the operator $\alpha(L)^{-1}$ sends any element in $\mathcal{M}$ to another random variable in $\mathcal{M}$. This implies that if the function $\alpha(z)$ has no roots in an open set containing the unit circle, we have the expression

$$\frac{1}{\alpha(z)} = \sum_{n=-\infty}^{\infty} a_n z^n,$$

which implies that we also have

$$\alpha(L)^{-1} = \sum_{n=-\infty}^{\infty} a_n L^n$$

in particular because (a_n) are absolutely summable. Similarly, if $\alpha(z)$ has no roots in an open set containing the unit disk $\{z : |z| \leq 1\}$, we have the Taylor expansion

$$\frac{1}{\alpha(z)} = \sum_{n=0}^{\infty} a_n z^n$$

and the associated filter

$$\alpha(L)^{-1} = \sum_{n=0}^{\infty} a_n L^n.$$

$\alpha(L)^{-1}$ having this one-sided expression is what people mean by the invertibility of AR processes.

2.1.1 Exercise

1. Suppose we have an AR(1) process given by

$$X_t = \alpha X_{t-1} + \varepsilon_t$$

Then, we may write

$$(I - \alpha L)X_t = \varepsilon_t$$

For $(1 - \alpha L)^{-1} = \sum_{n=0}^{\infty} (\alpha L)^n$ to be well defined, we need $\alpha(z)^{-1}$ to be analytic in an open set containing unit circle. This means all the roots of $\alpha(z) = 1 - \alpha z$ should be outside of the unit circle. The root is given by $z^* = \frac{1}{\alpha}$. Thus, we need $|\alpha| < 1$.

2. Suppose we have AR(2) process given by

$$X_t = \alpha_1 X_{t-1} + \alpha_2 X_{t-2} + \varepsilon_t$$

Then,

$$(I - \alpha_1 L - \alpha_2 L^2)X_t = \varepsilon_t$$

We need $\alpha(z)^{-1} = \frac{1}{1 - \alpha_1 z - \alpha_2 z^2}$ to be analytic inside the unit circle. This means the root of the function $\alpha(z) = 1 - \alpha_1 z - \alpha_2 z^2$ should be outside of the unit circle.

2.2 Different approach to invertibility of AR processes

Suppose we have an AR(2) process given by

$$X_t = \alpha_1 X_{t-1} + \alpha_2 X_{t-2} + \varepsilon_t$$

This has a companion VAR(1) form given by

$$\begin{pmatrix} X_t \\ X_{t-1} \end{pmatrix} = \begin{pmatrix} \alpha_1 & \alpha_2 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} X_{t-1} \\ X_{t-2} \end{pmatrix} + \begin{pmatrix} \varepsilon_t \\ 0 \end{pmatrix}$$

Repeated substitution gives

$$Y_t = e_t + A e_{t-1} + A^2 e_{t-2} + \dots \quad (5)$$

where

$$A = \begin{pmatrix} \alpha_1 & \alpha_2 \\ 1 & 0 \end{pmatrix}, Y_t = \begin{pmatrix} X_t \\ X_{t-1} \end{pmatrix}, e_t = \begin{pmatrix} \varepsilon_t \\ 0 \end{pmatrix}$$

For the RHS of (5) to converge, we need A^n to go to zero as $n \rightarrow \infty$. Let Jordan normal form of A be given by

$$A = Q\Lambda Q^{-1}$$

You can think of this as a generalization of eigendecomposition that works even if A is not diagonalizable. We have

$$A^n = Q\Lambda^n Q^{-1}$$

which implies that we need the eigenvalues of A to be strictly less than 1 for A^n to converge to zero. By definition, the eigenpairs (λ, v) are defined by

$$(A - \lambda I)v = 0$$

where $v \neq 0$. Thus, we need the determinant of

$$A - \lambda I = \begin{pmatrix} \alpha_1 - \lambda & \alpha_2 \\ 1 & -\lambda \end{pmatrix}$$

to be zero for λ to be an eigenvalue. This implies that if λ is an eigenvalue, we have

$$\lambda^2 - \alpha_1\lambda - \alpha_2 = 0 \tag{6}$$

Since we need the eigenvalues to be strictly less than 1, we need (6) to not have any roots inside of the unit circle. This is equivalent to requiring the roots of the function $\alpha(z) = 1 - \alpha_1z - \alpha_2z^2$ to be outside of the unit circle. This argument also generalizes to AR(p) processes. And also VAR(p) processes.